THE UNIVERSITY OF MELBOURNE

Student number

Semester 1 Assignment 1, 2025

School of Mathematics and Statistics

MAST20009 Vector Calculus

Submission deadline: 5pm on Monday 24 March 2025

This assignment consists of 10 pages (including this page) with 3 questions and 40 total marks

Instructions to Students

- This assignment is worth 5% of your total MAST20009 mark.
- If you have a printer, print the assignment one-sided.

Writing

- Assignments must be neatly written in blue or black pen or may be typeset using LaTeX.
- Full working must be shown in your solutions.
- Marks will be deducted for incomplete working, insufficient justification of steps, incorrect mathematical notation and for messy presentation of solutions.
- If you are writing answers on the assignment and you need more space, use blank paper. Note this in the answer box, so the marker knows.
- If you are only writing on blank A4 paper, the first page must contain only your student number, subject code and subject name. Write on one side of each sheet only. Start each question on a new page and include the question number at the top of each page.

Scanning and Submitting

- Put the pages in number order and the correct way up. Add any extra pages to the end. Use a scanning app to scan all pages to PDF. Scan directly from above. Crop pages to A4.
- Submit your scanned assignment as a single PDF file and carefully review the submission in Gradescope. Scan again and resubmit if necessary.

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Question 1 (15 marks)

Let

$$f(x,y) = \begin{cases} \dfrac{e^{(x+y)^2} - 1}{x^2 + y^2} & \text{if } (x,y) \neq (0,0) \\ 1 & \text{if } (x,y) = (0,0). \end{cases}$$

(a) Evaluate the limit of $f(x,y)$ when $(x,y) \to (0,0)$ along the lines $x = 0$, $y = 0$, and $y = kx$ for $k \in \mathbb{R} \setminus \{0\}$.

Along $x = 0$ the limit is

$$\begin{aligned} \lim_{y\to 0} \frac{e^{y^2} - 1}{y^2} &= \lim_{y\to 0} \frac{2ye^{y^2}}{2y} && \text{by L'Hopital's rule } (\tfrac{0}{0}) \\ &= \lim_{y\to 0} e^{y^2} \\ &= 1. \end{aligned}$$

Along $y = 0$ the limit is

$$\begin{aligned} \lim_{x\to 0} \frac{e^{x^2} - 1}{x^2} &= \lim_{x\to 0} \frac{2xe^{x^2}}{2x} && \text{by L'Hopital's rule } (\tfrac{0}{0}) \\ &= \lim_{x\to 0} e^{x^2} \\ &= 1. \end{aligned}$$

Along $y = kx$ the limit is

$$\begin{aligned} \lim_{x\to 0} \frac{e^{(k+1)^2x^2} - 1}{x^2 + k^2x^2} &= \lim_{x\to 0} \frac{2x(k+1)^2e^{(k+1)^2x^2}}{2x(k^2+1)} && \text{by L'Hopital's rule } (\tfrac{0}{0}) \\ &= \lim_{x\to 0} \frac{(k+1)^2e^{(k+1)^2x^2}}{(k^2+1)} \\ &= \frac{(k+1)^2}{k^2+1} \end{aligned}$$

(b) Determine where $f(x, y)$ is continuous. Justify your answer.

$f(x, y)$ is continuous for all $(x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$ by Theorems 3.2 and 3.3 from the lecture notes.

We get different limits when we approach $(0, 0)$ along different paths. Thus the limit at $(0, 0)$ does not exist. So f is not continuous at $(0, 0)$.

Hence $f(x, y)$ is continuous exactly when $(x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$.

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(c) Compute the partial derivative $\frac{\partial f}{\partial x}$ at $(x, y) = (0, 0)$.

We can compute $f_x(x, y)$ for general (x, y) and then attempt to substitute $(0, 0)$, but we might guess that this won't work because of the $x^2 + y^2$ in the denominator of $f(x, y)$.

Instead we can write

$$\begin{aligned}
f_x(0,0) &= \lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h} \\
&= \lim_{h\to 0} \frac{\frac{e^{h^2}-1}{h^2} - 1}{h} \\
&= \lim_{h\to 0} \frac{e^{h^2} - h^2 - 1}{h^3} \\
&= \lim_{h\to 0} \frac{2he^{h^2} - 2h}{3h^2} && \text{by L'Hopital's rule } (\tfrac{0}{0}) \\
&= \lim_{h\to 0} \frac{2e^{h^2} + 4h^2e^{h^2} - 2}{6h} && \text{by L'Hopital's rule } (\tfrac{0}{0}) \\
&= \lim_{h\to 0} \frac{4he^{h^2} + 8he^{h^2} + 8h^3e^{h^2}}{6} && \text{by L'Hopital's rule } (\tfrac{0}{0}) \\
&= 0.
\end{aligned}$$

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Question 2 (10 marks)

Let the functions $f : \mathbb{R}^2 \to \mathbb{R}^2$ and $g : \mathbb{R}^3 \to \mathbb{R}^2$ be respectively

$$f(u, v) = (uv + u + v, u \cos v)$$
$$g(x, y, z) = (x + y + x \sinh z, xe^{x+z} + 2xy).$$

(a) Find the derivative matrices $\mathbf{D}f$ and $\mathbf{D}g$.

$$\mathbf{D}f = \begin{bmatrix} v+1 & u+1 \\ \cos v & -u \sin v \end{bmatrix}$$
$$\mathbf{D}g = \begin{bmatrix} 1 + \sinh z & 1 & x \cosh z \\ (x+1)e^{x+z} + 2y & 2x & xe^{x+z} \end{bmatrix}$$

(b) Find $\mathbf{D}(f \circ g)$ at $(x, y, z) = (0, 2, 0)$.

At $(x, y, z) = (0, 2, 0)$, $g(x, y, z) = (2, 0)$. By the chain rule we have

$$\begin{aligned}\mathbf{D}(f \circ g)(0, 2, 0) &= \mathbf{D}f(2, 0)\mathbf{D}g(0, 2, 0) \\ &= \begin{bmatrix} 1 & 3 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 5 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 16 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}\end{aligned}$$

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Question 3 (15 marks)

Let $f(x, y) = (x + 1)^2 \cos(2y)$.

(a) Determine the first order Taylor polynomial $p_1(x, y)$ for f at $(0, 0)$ and write down the remainder $R_1(x, y)$. Note that your expression for $R_1(x, y)$ should involve a parameter ξ as used in lectures. For this part you do not need to explicitly write down what f_{xx}, f_{xy}, and f_{yy} are (though you will need them for the next question part).

We first have

$$f_x(x, y) = 2(x + 1)\cos(2y)$$
$$f_y(x, y) = -2(x + 1)^2 \sin(2y)$$

so

$$f(0, 0) = 1, \qquad f_x(0, 0) = 2, \qquad f_y(0, 0) = 0.$$

Hence

$$p_1(x, y) = 1 + 2x.$$

Then

$$R_1(x, y) = \frac{x^2}{2!} f_{xx}(\xi x, \xi y) + \frac{xy}{1!1!} f_{xy}(\xi x, \xi y) + \frac{y^2}{2!} f_{yy}(\xi x, \xi y)$$

for some $0 < \xi < 1$.

(b) Determine the second order Taylor polynomial $p_2(x, y)$ for f at $(0, 0)$.

We have

$$\begin{aligned} f_{xx}(x, y) &= 2\cos(2y) \\ f_{xy}(x, y) &= -4(x+1)\sin(2y) \\ f_{yy}(x, y) &= -4(x+1)^2\cos(2y) \end{aligned}$$

so

$$f_{xx}(0, 0) = 2, \qquad f_{xy}(0, 0) = 0, \qquad f_{yy}(0, 0) = -4.$$

Then (using the answer from (a)),

$$p_2(x, y) = 1 + 2x + \frac{2x^2}{2!} - \frac{4y^2}{2!} = 1 + 2x + x^2 - 2y^2.$$

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(c) Estimate $f(0.1, -0.1)$ using p_1. Repeat this using p_2. Also compute the exact value (to a few decimal places), to verify that your approximations become more accurate as you include more terms.

$$p_1(0.1, -0.1) = 1 + 2(0.1) = 1.2$$

$$p_2(0.1, -0.1) = 1 + 2(0.1) + (0.1)^2 - 2(-0.1)^2 = 1.19$$

$$f(0.1, -0.1) = (0.1 + 1)^2 \cos(2 \times 0.1) \approx 1.18588$$

End of Assignment — Total Available Marks = 40